

LAB4: SUBNETTING AND SUPERNETTING USING CISCO PACKET TRACER

✓ **OBJECTIVES:**

- To understand the concept of subnetting and super netting
- To learn how to efficiently manage IP addresses
- To implement subnetting and supernetting using Cisco Packet Tracer
- To analyze how these techniques improve routing and network performance

✓ **TOOLS AND SOFTWARE USED**

- Cisco Packet Tracer
- PC with Windows/Linux OS

✓ **THOERY**

The term associated with this lab are given below:

1. Subnetting

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets. It helps to efficiently utilize IP addresses, reduce network congestion, and improve security and performance.

Working of Subnetting

Subnetting works by borrowing bits from the host portion of an IP address and adding them to the network portion. This increases the number of network IDs while reducing the number of hosts available in each subnet. As a result, each subnet has its own unique network address, broadcast address, and valid host range. This structure allows better control over the network, efficient management of IP addresses, and reduced network traffic..

Applications of Subnetting

- Used in organizations to divide networks by departments (HR, IT, Finance, etc.)
- Helps ISPs allocate IP addresses efficiently
- Improves network security by isolating subnets
- Reduces broadcast traffic in large networks
- Commonly used in enterprise and campus networks

2. Supernetting

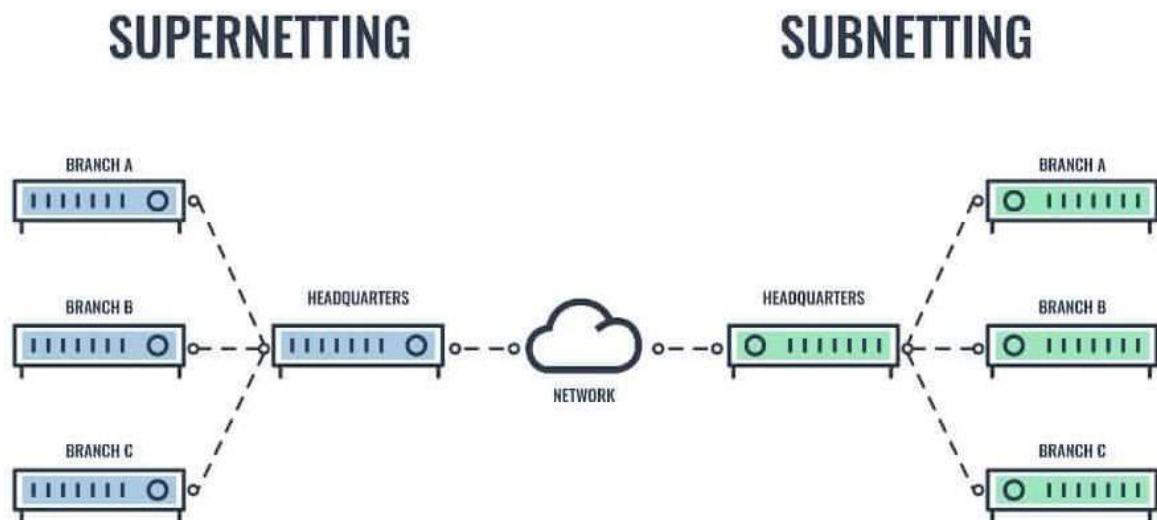
Supernetting is the process of combining multiple smaller IP networks into a single larger logical network called a supernet. It helps to reduce routing table size, improve routing efficiency, and simplify network management.

Working of Supernetting

Supernetting works by borrowing bits from the network portion of an IP address and adding them to the host portion. This process reduces the number of network IDs while increasing the number of hosts in the combined network. By summarizing multiple contiguous networks into a single route using a shorter subnet mask, supernetting enables efficient routing and reduces routing overhead.

Applications of Supernetting

- Used in route summarization to reduce routing table size
- Improves routing performance in large networks
- Helps ISPs manage and advertise multiple networks efficiently
- Reduces routing overhead and processing time
- Commonly used in backbone and wide area networks (WANs)



✓ **PROCEDURE**

✓ Subnetting in Cisco Packet Tracer

- Open Cisco Packet Tracer
- Add routers, switches, and PCs
- Assign IP addresses according to the subnet table
- Configure router interfaces with appropriate IP addresses
- Set default gateway on each PC
- Verify connectivity using ping command

✓ Super-netting in Cisco Packet Tracer

- Create multiple networks as per given IP ranges
- Configure routing using static or dynamic routing
- Route summarization using supernet address
- Verify routing table and connectivity

✓ NETWORK DESIGN

Calculation of subnet :

Base network: 192.168.1.0/24

Required number of subnet: 4

Number of IP address per subnet : 64 (Block size)

/26 (Borrowed 2 bits: $2^2 = 4$ subnets)

Subnet mask: 255.255.255.192 (11111111 . 11111111 . 11111111 . 11000000)

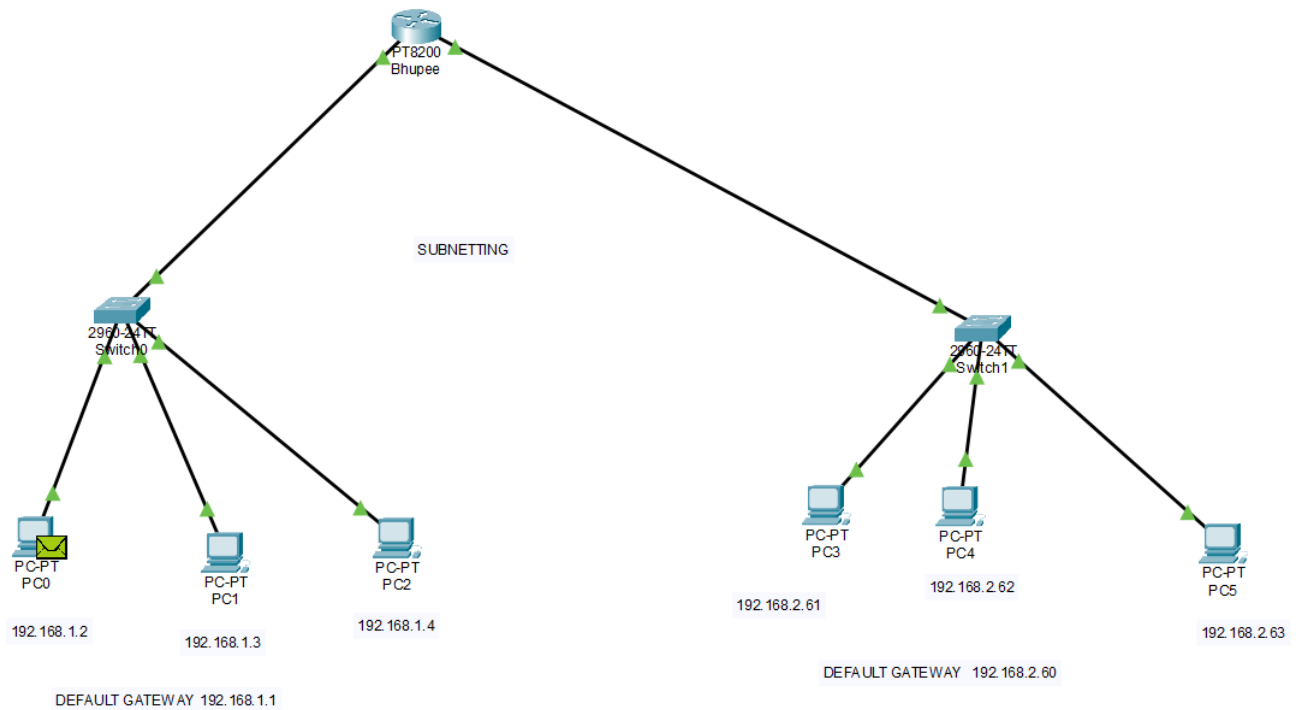
Subnet calculation Table

<u>Subnet</u>	<u>Network ID</u>	<u>Broadcast Address</u>	<u>1st usable IP</u>	<u>Last Usable IP</u>
<u>1</u>	<u>192.168.1.0</u>	<u>192.168.1.63</u>	<u>192.168.1.1</u>	<u>192.168.1.62</u>
<u>2</u>	<u>192.168.1.64</u>	<u>192.168.1.127</u>	<u>192.168.1.65</u>	<u>192.168.1.126</u>
<u>3</u>	<u>192.168.1.128</u>	<u>192.168.1.191</u>	<u>192.168.1.129</u>	<u>192.168.1.190</u>
<u>4</u>	<u>192.168.1.192</u>	<u>192.168.1.255</u>	<u>192.168.1.193</u>	<u>192.168.1.254</u>

Network Topology

Subnetting:

A network was set up using Router0 to link two subnets: 192.168.1.0/26 and 192.168.1.64/26. Switch0 connects PC0, PC1, and PC2 to the first subnet, while Switch1 connects PC3, PC4, and PC5 to the second. Both switches are linked to the router's interfaces.

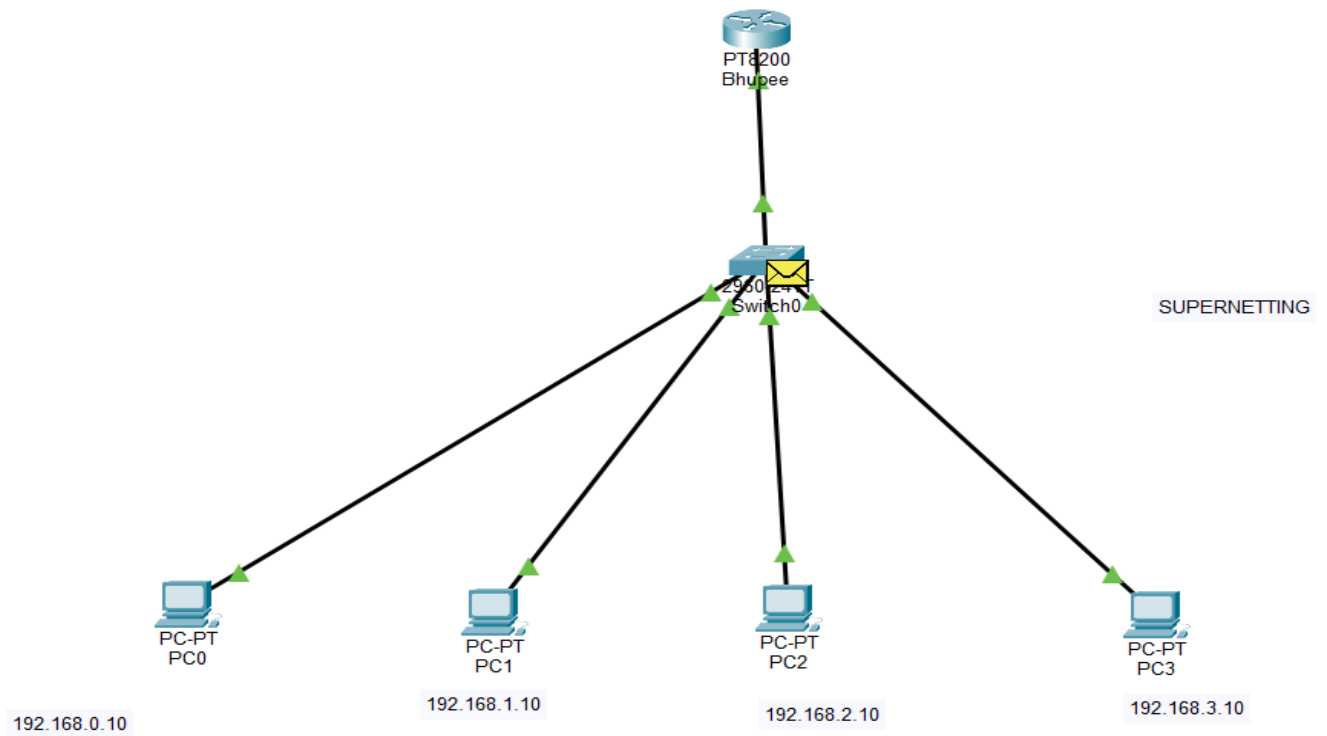


Configuration table Subnetting

Device	IPV4	Subnet-mask	Default gateway
Router0 (FastEthernet0/0)	192.168.1.1	255.255.255.192	N/A
Router0 (FastEthernet0/1)	192.168.2.60	255.255.255.192	N/A
Pc0(Subnet 1)	192.168.1.2	255.255.255.192	192.168.1.1
Pc1(Subnet 1)	192.168.1.3	255.255.255.192	192.168.1.1
Pc2(Subnet 1)	192.168.1.4	255.255.255.192	192.168.1.1
Pc3(Subnet 1)	192.168.2.61	255.255.255.192	192.168.2.60
Pc4(Subnet 1)	192.168.2.62	255.255.255.192	192.168.2.60
Pc5(Subnet 1)	192.168.2.63	255.255.255.192	192.168.2.60

Supernetting:

The network has one router, one switch, and four PCs (PC0–PC3). All PCs connect to the switch, which connects to the router.



Configuration Table:

<u>Device</u>	<u>IPv4</u>	<u>Subnet-mask</u>	<u>Default gateway</u>
Router0 (FastEthernet0/0)	192.168.0.1	255.255.252.0	N/A
PC0	192.168.0.10	255.255.252.0	192.168.0.1
PC1	192.168.1.10	255.255.252.0	192.168.0.1
PC2	192.168.2.10	255.255.252.0	192.168.0.1
PC3	192.168.3.10	255.255.252.0	192.168.0.1

Ping from PC0 to PC3

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

✓ **PRECAUTION**

1. Double-check IP addresses and subnet masks before assigning them to devices.
2. Ensure routers are properly configured to allow communication between subnets.
3. Avoid overlapping IP addresses when subnetting or supernetting.
4. Test connectivity using ping or other diagnostic commands after configuration.
5. Save your Packet Tracer project frequently to prevent loss of work.

✓ **DISCUSSION**

Subnetting is the process of dividing a large network into smaller sub-networks to improve performance, security, and efficient use of IP addresses. It helps reduce network traffic and makes management easier. Supernetting is the opposite process, where multiple smaller networks are combined into one larger network to simplify routing. This reduces routing table size and improves scalability. In Cisco Packet Tracer, subnetting and supernetting can be implemented by configuring IP addresses, subnet masks, and routing protocols. Connectivity can be verified using the ping command to ensure proper communication between devices.

✓ **RESULT**

After setting up the subnetted networks in Packet Tracer, devices within the same subnet communicated without any issues. Communication between different subnets was possible only after configuring a router and enabling proper routing. Once routing was correctly set, data transfer between subnets worked smoothly. In the case of supernetting, combining smaller networks helped reduce routing table entries and simplified network management. Testing with the ping command and verifying the routing table showed that the network configuration was successful and functioning properly.

✓ **CONCLUSION**

Subnetting divides a large network into smaller, manageable parts to improve efficiency, security, and performance. In contrast, supernetting merges smaller networks into a larger one to reduce routing complexity and simplify network management. Cisco Packet Tracer provides a practical environment to design, configure, and test these networking concepts. Through hands-on practice, users gain a clearer understanding of IP addressing, routing, and overall network structure. Proper configuration of subnetting and supernetting leads to better network organization and smoother communication.